

Machine Learning: A Comprehensive Textbook

Solutions to Chapter 4 Exercises

Regression: From Simple Lines to Regularised Models

Solutions Manual

Exercise 4.1 — Normal Equations from Scratch

Starting from $J(\theta) = \frac{1}{2n} \|y - X\theta\|_2^2$, derive the Normal Equations using $\nabla_\theta(a^\top \theta) = a$ and $\nabla_\theta(\theta^\top A \theta) = 2A\theta$ ($A = A^\top$). State when the solution is unique.

Solution. This mirrors Exercise 2.4 with an extra $\frac{1}{n}$ factor. Expand:

$$J(\theta) = \frac{1}{2n} (y - X\theta)^\top (y - X\theta) = \frac{1}{2n} (y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X \theta).$$

Applying the given identities term by term (with $A = X^\top X = A^\top$):

$$\nabla_\theta J = \frac{1}{2n} (-2X^\top y + 2X^\top X \theta) = \frac{1}{n} (X^\top X \theta - X^\top y).$$

Setting $\nabla_\theta J = 0$:

$$X^\top X \theta = X^\top y \quad (\text{the Normal Equations}).$$

Uniqueness. The Hessian is $\nabla_\theta^2 J = \frac{1}{n} X^\top X \succeq 0$ (always PSD), so J is convex and any stationary point is a global minimum. The solution $\theta^* = (X^\top X)^{-1} X^\top y$ is *unique* if and only if $X^\top X$ is invertible, which holds iff X has full column rank d (i.e. the d columns/features of X are linearly independent and $n \geq d$). If $X^\top X$ is singular (e.g. $n < d$, or exact multicollinearity as in Exercise 4.7), the Normal Equations still have solutions (since $X^\top y$ lies in the row space/range of $X^\top X$), but there are infinitely many — a whole affine subspace of minimizers — and one typically selects the minimum-norm solution via the pseudoinverse, $\theta^* = (X^\top X)^\dagger X^\top y$.

Exercise 4.2 — Probabilistic Derivation

Show that maximising the log-likelihood under $y^{(i)} = \theta^\top x^{(i)} + \epsilon^{(i)}$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$, is equivalent to minimising $\|y - X\theta\|_2^2$.

Solution. Under the model, $y^{(i)} | x^{(i)}; \theta \sim \mathcal{N}(\theta^\top x^{(i)}, \sigma^2)$, so the likelihood of the n i.i.d. observations is

$$L(\theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y^{(i)} - \theta^\top x^{(i)})^2}{2\sigma^2}\right).$$

Taking the log:

$$\ell(\theta) = \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y^{(i)} - \theta^\top x^{(i)})^2}{2\sigma^2} \right] = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y^{(i)} - \theta^\top x^{(i)})^2.$$

The first term does not depend on θ , and $\sigma^2 > 0$ is a fixed constant, so

$$\arg \max_{\theta} \ell(\theta) = \arg \max_{\theta} \left[-\frac{1}{2\sigma^2} \sum_i (y^{(i)} - \theta^\top x^{(i)})^2 \right] = \arg \min_{\theta} \sum_i (y^{(i)} - \theta^\top x^{(i)})^2 = \arg \min_{\theta} \|y - X\theta\|_2^2.$$

Hence MLE under Gaussian-noise linear regression is exactly equivalent to ordinary least squares — least squares is the MLE precisely because squared error is (up to constants) the negative log-density of a Gaussian.

Exercise 4.3 — OLS is Unbiased

Prove $\mathbb{E}[\hat{\theta}_{\text{OLS}}] = \theta^*$ under the classical linear model. What assumption is crucial?

Solution. Assume the true model is $y = X\theta^* + \epsilon$ with $\mathbb{E}[\epsilon \mid X] = 0$ (the crucial exogeneity/strict-exogeneity assumption — errors are mean-zero *conditional on the design matrix* X), and X has full column rank so $(X^\top X)^{-1}$ exists.

The OLS estimator is $\hat{\theta}_{\text{OLS}} = (X^\top X)^{-1} X^\top y$. Substitute the true model:

$$\hat{\theta}_{\text{OLS}} = (X^\top X)^{-1} X^\top (X\theta^* + \epsilon) = (X^\top X)^{-1} X^\top X\theta^* + (X^\top X)^{-1} X^\top \epsilon = \theta^* + (X^\top X)^{-1} X^\top \epsilon.$$

Taking the conditional expectation given X :

$$\mathbb{E}[\hat{\theta}_{\text{OLS}} \mid X] = \theta^* + (X^\top X)^{-1} X^\top \mathbb{E}[\epsilon \mid X] = \theta^* + (X^\top X)^{-1} X^\top \cdot 0 = \theta^*.$$

By the law of total expectation, $\mathbb{E}[\hat{\theta}_{\text{OLS}}] = \mathbb{E}_X[\mathbb{E}[\hat{\theta}_{\text{OLS}} \mid X]] = \mathbb{E}_X[\theta^*] = \theta^*$. ■

Crucial assumption. $\mathbb{E}[\epsilon \mid X] = 0$ — the errors must be uncorrelated with (more strongly, mean-independent of) the regressors. If this fails (e.g. due to omitted-variable bias, measurement error in X , or simultaneity/endogeneity), $\mathbb{E}[(X^\top X)^{-1} X^\top \epsilon] \neq 0$ in general, and OLS becomes biased even asymptotically (inconsistent), which is the classical motivation for instrumental-variables methods in econometrics.

Exercise 4.4 — Ridge Shrinkage via SVD

Using $X = U\Sigma V^\top$, show Ridge predictions are $\hat{y}_{\text{Ridge}} = \sum_j \frac{\sigma_j^2}{\sigma_j^2 + \lambda} u_j u_j^\top y$. Interpret each shrinkage factor.

Solution. The Ridge estimator is $\hat{\theta}_{\text{Ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$, and predictions are $\hat{y} = X\hat{\theta}_{\text{Ridge}}$. Substitute $X = U\Sigma V^\top$ (with $U \in \mathbb{R}^{n \times d}$, $\Sigma \in \mathbb{R}^{d \times d}$ diagonal, $V \in \mathbb{R}^{d \times d}$ orthogonal, assuming $n \geq d$, thin SVD):

$$X^\top X = V\Sigma U^\top U\Sigma V^\top = V\Sigma^2 V^\top \text{ (using } U^\top U = I \text{). So}$$

$$X^\top X + \lambda I = V\Sigma^2 V^\top + \lambda VV^\top = V(\Sigma^2 + \lambda I)V^\top,$$

using $VV^\top = I$ (since V is square orthogonal). Hence

$$(X^\top X + \lambda I)^{-1} = V(\Sigma^2 + \lambda I)^{-1} V^\top.$$

Then

$$\hat{y}_{\text{Ridge}} = X\hat{\theta}_{\text{Ridge}} = U\Sigma V^\top \cdot V(\Sigma^2 + \lambda I)^{-1} V^\top \cdot V\Sigma U^\top y = U\Sigma(\Sigma^2 + \lambda I)^{-1} \Sigma U^\top y,$$

using $V^\top V = I$ repeatedly. Since $\Sigma, (\Sigma^2 + \lambda I)^{-1}$ are both diagonal, $\Sigma(\Sigma^2 + \lambda I)^{-1} \Sigma$ is diagonal with j -th entry $\sigma_j^2/(\sigma_j^2 + \lambda)$. Writing U 's columns as u_j :

$$\hat{y}_{\text{Ridge}} = \sum_{j=1}^d \frac{\sigma_j^2}{\sigma_j^2 + \lambda} u_j u_j^\top y. \quad \blacksquare$$

(Compare to OLS, $\hat{y}_{\text{OLS}} = \sum_j u_j u_j^\top y$, i.e. shrinkage factor exactly 1 for every direction.)

Interpretation. Each principal direction u_j of X (ranked by singular value σ_j , i.e. by how much variance X has along that direction) contributes to the fitted $\hat{y}$ with a shrinkage factor $\sigma_j^2/(\sigma_j^2 + \lambda) \in (0, 1)$:

- For directions with *large* σ_j (well-determined, high-variance directions in X), the shrinkage factor is close to 1 — Ridge barely shrinks these, because the data strongly constrains them and overfitting risk is low.
- For directions with *small* σ_j (poorly-determined, low-variance/near-collinear directions), the shrinkage factor is close to 0 — Ridge aggressively shrinks these toward zero, precisely because these are the directions where OLS coefficients would otherwise blow up (small σ_j in the denominator of $(X^\top X)^{-1} \propto 1/\sigma_j^2$ makes OLS extremely sensitive to noise along that direction).

This is exactly how Ridge trades bias for variance: it leaves well-estimated directions alone and stabilizes poorly-estimated (high-variance) directions.

Exercise 4.5 — LASSO Soft-Thresholding

For orthogonal-design LASSO $\min_\theta \frac{1}{2}(\hat{\theta}^{\text{OLS}} - \theta)^2 + \lambda|\theta|$, derive $\hat{\theta} = S_\lambda(\hat{\theta}^{\text{OLS}})$ using subgradients.

Solution. Let $f(\theta) = \frac{1}{2}(\hat{\theta}^{\text{OLS}} - \theta)^2 + \lambda|\theta|$, $\lambda > 0$. Write $a := \hat{\theta}^{\text{OLS}}$ for brevity. The subdifferential of $|\theta|$ is

$$\partial|\theta| = \begin{cases} \{1\} & \theta > 0 \\ \{-1\} & \theta < 0 \\ [-1, 1] & \theta = 0. \end{cases}$$

The (sub)gradient optimality condition is $0 \in \partial f(\theta) = -(a - \theta) + \lambda \partial|\theta|$, i.e. $(\theta - a) + \lambda \partial|\theta| \ni 0$.

Case $\theta > 0$: $\partial|\theta| = \{1\}$, so the condition is $\theta - a + \lambda = 0 \implies \theta = a - \lambda$. This is a valid solution in this case only if $a - \lambda > 0$, i.e. $a > \lambda$.

Case $\theta < 0$: $\partial|\theta| = \{-1\}$, so $\theta - a - \lambda = 0 \implies \theta = a + \lambda$. Valid only if $a + \lambda < 0$, i.e. $a < -\lambda$.

Case $\theta = 0$: need $0 \in -a + \lambda[-1, 1]$, i.e. $a \in [-\lambda, \lambda]$, i.e. $|a| \leq \lambda$.

These three cases are mutually exclusive and exhaustive over all values of a (any a falls into exactly one of $a > \lambda$, $a < -\lambda$, $|a| \leq \lambda$), giving the closed-form solution:

$$\hat{\theta} = S_\lambda(a) = \begin{cases} a - \lambda & a > \lambda \\ 0 & |a| \leq \lambda \\ a + \lambda & a < -\lambda \end{cases} = \text{sign}(a) \max(|a| - \lambda, 0),$$

the **soft-thresholding operator**. Geometrically: LASSO shrinks every OLS coefficient toward zero by a fixed amount λ , and coefficients with $|\hat{\theta}^{\text{OLS}}| \leq \lambda$ are set *exactly* to zero — this is the mechanism behind LASSO's sparsity-inducing property, in contrast to Ridge's proportional (never-exactly-zero) shrinkage from Exercise 4.4.

Exercise 4.6 — Bayesian Perspective on Regularisation

(a) Ridge = MAP with $\theta \sim \mathcal{N}(0, \sigma^2 \lambda^{-1} I)$. (b) LASSO = MAP with Laplace prior. (c) What prior gives ℓ_0 ?

Solution. (a) Ridge. This generalizes Exercise 2.7 to the regression (Gaussian-likelihood) setting. With likelihood $y^{(i)} \mid \theta \sim \mathcal{N}(\theta^\top x^{(i)}, \sigma^2)$ and prior $\theta \sim \mathcal{N}(0, \sigma^2 \lambda^{-1} I)$:

$$\log p(\theta \mid \mathcal{D}) \stackrel{+c}{=} -\frac{1}{2\sigma^2} \|y - X\theta\|_2^2 - \frac{\lambda}{2\sigma^2} \|\theta\|_2^2,$$

(the prior log-density is $-\frac{1}{2\sigma^2\lambda} \|\theta\|_2^2 = -\frac{\lambda}{2\sigma^2} \|\theta\|_2^2$ up to a θ -independent constant). Maximizing the posterior is equivalent to

$$\arg \min_{\theta} \left[\|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2 \right],$$

after multiplying through by $2\sigma^2$ — exactly the Ridge objective with penalty coefficient λ .

(b) LASSO. With Laplace prior $p(\theta_j) \propto \exp(-\lambda|\theta_j|/\sigma^2)$ i.i.d. across coordinates, $\log p(\theta) \stackrel{+c}{=} -\frac{\lambda}{\sigma^2} \sum_j |\theta_j| = -\frac{\lambda}{\sigma^2} \|\theta\|_1$. Combined with the same Gaussian likelihood:

$$\log p(\theta \mid \mathcal{D}) \stackrel{+c}{=} -\frac{1}{2\sigma^2} \|y - X\theta\|_2^2 - \frac{\lambda}{\sigma^2} \|\theta\|_1.$$

Maximizing the posterior (equivalently minimizing the negative, times $2\sigma^2$) gives

$$\arg \min_{\theta} \left[\|y - X\theta\|_2^2 + 2\lambda \|\theta\|_1 \right],$$

which is exactly the LASSO objective (with penalty 2λ , an inconsequential rescaling of the hyperparameter). The Laplace prior's sharp peak at 0 (relative to a Gaussian) is what induces exact sparsity in the MAP estimate, mirroring the corner/non-smoothness of the ℓ_1 ball.

(c) ℓ_0 prior. ℓ_0 regularisation penalizes the *number* of nonzero coefficients, $\|\theta\|_0 = \sum_j \mathbf{1}[\theta_j \neq 0]$. The corresponding prior would need $\log p(\theta) \propto -\lambda \|\theta\|_0$, i.e. a mixture (“spike-and-slab”) prior: each coefficient is, independently, exactly zero with some probability π (a point mass / “spike” at 0) or drawn from a diffuse “slab” distribution (e.g. a flat or wide Gaussian) with probability $1 - \pi$. Formally, $p(\theta_j) = \pi \delta_0(\theta_j) + (1 - \pi) p_{\text{slab}}(\theta_j)$; the MAP estimate under this prior directly penalizes the count of active coefficients (each active, nonzero coefficient incurs a fixed $-\log \frac{1-\pi}{\pi}$ cost relative to being zero, matching ℓ_0 's per-coefficient penalty), unlike Laplace (LASSO), which instead penalizes coefficient *magnitude* continuously.

Exercise 4.7 — Multicollinearity

Two features perfectly correlated, $x_2^{(i)} = x_1^{(i)}$ for all i . Show $X^\top X$ is singular. Describe geometrically what happens and how Ridge resolves it.

Solution. Singularity. Let $X = [x_1 \ x_2 \ \cdots]$ with $x_2 = x_1$ (columns identical). Then $X^\top X$ has identical first two rows/columns (since $(X^\top X)_{1k} = x_1^\top x_k$ and $(X^\top X)_{2k} = x_2^\top x_k = x_1^\top x_k$ for all k , using $x_2 = x_1$). A matrix with two identical rows has determinant 0 (row operations: subtracting row 2 from row 1 gives an all-zero row), hence $X^\top X$ is singular (not invertible). Equivalently: the vector $v = (1, -1, 0, \dots, 0)^\top$ satisfies $Xv = x_1 - x_2 = 0$, so X does not have full column rank, hence $\text{rank}(X^\top X) = \text{rank}(X) < d$.

Geometric interpretation. For any solution $\theta^* = (\theta_1^*, \theta_2^*, \dots)$ of the Normal Equations and any $c \in \mathbb{R}$, $\theta' = (\theta_1^* + c, \theta_2^* - c, \dots)$ is *also* a solution, because $X\theta' = \theta_1^* x_1 + cx_1 + \theta_2^* x_2 - cx_2 + \cdots = X\theta^*$ (the c terms cancel exactly since $x_1 = x_2$). So the fitted *predictions* $\hat{y} = X\theta$ are perfectly well-defined and unique (there's a unique projection of y onto the column space of X), but the individual

coefficients θ_1, θ_2 are not identifiable — only their sum $\theta_1 + \theta_2$ matters, and there is an entire line (a 1-dimensional affine subspace) of θ -values all giving the same predictions. This is the classic symptom of multicollinearity: OLS coefficient estimates become arbitrary/unstable (in practice, with near- rather than exact-collinearity, this manifests as huge coefficient variance and sign flips under small data perturbations).

How Ridge resolves it. Ridge adds λI to $X^\top X$ before inverting: $(X^\top X + \lambda I)$ is now strictly positive definite (even though $X^\top X$ is only PSD with a zero eigenvalue along the $v = (1, -1, 0, \dots)$ direction, adding λI shifts that eigenvalue to $\lambda > 0$), so the inverse exists and the solution is unique. Among the entire line of OLS-equivalent solutions, Ridge's ℓ_2 penalty specifically selects the *minimum-norm* point on that line — by symmetry (since x_1, x_2 contribute identically), Ridge assigns $\theta_1 = \theta_2 = (\theta_1^* + \theta_2^*)/2$, splitting the “credit” for the shared predictive signal equally between the two collinear features, rather than letting either coefficient blow up in magnitude.

Exercise 4.8 — Bias-Variance for Ridge

Let $A_\lambda = (X^\top X + \lambda I)^{-1} X^\top X$. (a) Show $\text{Bias}(\hat{\theta}_{\text{Ridge}}) = (A_\lambda - I)\theta^*$. (b) Show $\text{Var}(\hat{\theta}_{\text{Ridge}}) \preceq \sigma^2(X^\top X)^{-1}$. (c) Prove $\exists \lambda^* > 0$ with $\text{MSE}(\hat{\theta}_{\text{Ridge}}(\lambda^*)) < \text{MSE}(\hat{\theta}_{\text{OLS}})$.

Solution. Write $\hat{\theta}_{\text{Ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$. Substituting $y = X\theta^* + \epsilon$, $\mathbb{E}[\epsilon] = 0$, $\text{Cov}(\epsilon) = \sigma^2 I$:

$$\hat{\theta}_{\text{Ridge}} = (X^\top X + \lambda I)^{-1} X^\top (X\theta^* + \epsilon) = \underbrace{(X^\top X + \lambda I)^{-1} X^\top X}_{A_\lambda} \theta^* + (X^\top X + \lambda I)^{-1} X^\top \epsilon.$$

(a) **Bias.** Taking expectation (over ϵ , conditional on X):

$$\mathbb{E}[\hat{\theta}_{\text{Ridge}}] = A_\lambda \theta^* + (X^\top X + \lambda I)^{-1} X^\top \mathbb{E}[\epsilon] = A_\lambda \theta^*.$$

So $\text{Bias}(\hat{\theta}_{\text{Ridge}}) = \mathbb{E}[\hat{\theta}_{\text{Ridge}}] - \theta^* = A_\lambda \theta^* - \theta^* = (A_\lambda - I)\theta^*$. ■ (Note: for $\lambda > 0$, $A_\lambda \neq I$ in general, so Ridge is biased — unlike OLS where $A_0 = I$.)

(b) **Variance.** Let $M_\lambda := (X^\top X + \lambda I)^{-1} X^\top$, so $\hat{\theta}_{\text{Ridge}} = A_\lambda \theta^* + M_\lambda \epsilon$ and

$$\text{Var}(\hat{\theta}_{\text{Ridge}}) = M_\lambda \text{Cov}(\epsilon) M_\lambda^\top = \sigma^2 M_\lambda M_\lambda^\top = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1},$$

using $M_\lambda^\top = X(X^\top X + \lambda I)^{-1}$ (the matrix $X^\top X + \lambda I$ is symmetric). Using the SVD-diagonalization from Exercise 4.4, $X^\top X = V \Sigma^2 V^\top$, so in the eigenbasis V , $\text{Var}(\hat{\theta}_{\text{Ridge}})$ is diagonalized with entries

$$\sigma^2 \cdot \frac{\sigma_j^2}{(\sigma_j^2 + \lambda)^2} \quad (j = 1, \dots, d),$$

while $\text{Var}(\hat{\theta}_{\text{OLS}}) = \sigma^2 (X^\top X)^{-1} = \sigma^2 V \Sigma^{-2} V^\top$ has (in the same eigenbasis) entries σ^2 / σ_j^2 . Comparing term by term (both are diagonal in the V -basis, so comparing diagonal entries suffices for the PSD ordering):

$$\frac{\sigma_j^2}{(\sigma_j^2 + \lambda)^2} \leq \frac{1}{\sigma_j^2} \iff \sigma_j^4 \leq (\sigma_j^2 + \lambda)^2 \iff \sigma_j^2 \leq \sigma_j^2 + \lambda,$$

which holds for all $\lambda \geq 0$ (with strict inequality for $\lambda > 0$). Hence every eigenvalue of $\sigma^2 (X^\top X)^{-1} - \text{Var}(\hat{\theta}_{\text{Ridge}})$ (in the shared eigenbasis V) is ≥ 0 , i.e. $\text{Var}(\hat{\theta}_{\text{Ridge}}) \preceq \sigma^2 (X^\top X)^{-1} = \text{Var}(\hat{\theta}_{\text{OLS}})$ in the PSD (Loewner) order. ■

(c) **There exists $\lambda^* > 0$ with lower MSE.** The total (scalar, trace) MSE decomposes as

$$\text{MSE}(\hat{\theta}_{\text{Ridge}}(\lambda)) = \|\text{Bias}(\hat{\theta}_{\text{Ridge}})\|_2^2 + \text{tr}(\text{Var}(\hat{\theta}_{\text{Ridge}})) =: B(\lambda)^2 + V(\lambda).$$

Consider $g(\lambda) := \text{MSE}(\hat{\theta}_{\text{Ridge}}(\lambda))$ as a function of $\lambda \geq 0$; $g(0) = \text{MSE}(\hat{\theta}_{\text{OLS}})$ (Ridge reduces to OLS at $\lambda = 0$, with zero bias). Differentiate at $\lambda = 0$:

- $B(\lambda)^2 = \theta^{*\top} (A_\lambda - I)^\top (A_\lambda - I) \theta^*$; at $\lambda = 0$, $A_0 = I$ so $B(0) = 0$, and $B(\lambda)^2 = O(\lambda^2)$ as $\lambda \rightarrow 0$ (since $A_\lambda - I = O(\lambda)$ — a first-order Taylor expansion of $(X^\top X + \lambda I)^{-1} X^\top X$ around $\lambda = 0$ shows the bias grows linearly in λ to first order, so its square is $O(\lambda^2)$), i.e. $\frac{d}{d\lambda} B(\lambda)^2|_{\lambda=0} = 0$.
- $V(\lambda) = \sigma^2 \sum_j \frac{\sigma_j^2}{(\sigma_j^2 + \lambda)^2}$; differentiating, $\frac{dV}{d\lambda} = \sigma^2 \sum_j \sigma_j^2 \cdot \frac{-2}{(\sigma_j^2 + \lambda)^3}$, which at $\lambda = 0$ is $-2\sigma^2 \sum_j \frac{1}{\sigma_j^4} < 0$ — strictly negative (assuming at least one $\sigma_j < \infty$, i.e. any real finite design, and no $\sigma_j = 0$, i.e. X full rank so OLS variance is finite).

Hence $g'(0) = \frac{d}{d\lambda} [B(\lambda)^2 + V(\lambda)]_{\lambda=0} = 0 + \left(-2\sigma^2 \sum_j \sigma_j^{-4}\right) < 0$. Since g is differentiable with strictly negative derivative at $\lambda = 0$, there exists $\lambda^* > 0$ small enough that $g(\lambda^*) < g(0)$, i.e.

$$\text{MSE}(\hat{\theta}_{\text{Ridge}}(\lambda^*)) < \text{MSE}(\hat{\theta}_{\text{OLS}}). \quad \blacksquare$$

This is the classical justification for Ridge regression: starting from $\lambda = 0$ (OLS), a small amount of bias always buys a strictly larger reduction in variance, so total MSE strictly decreases for sufficiently small $\lambda > 0$ (as long as X has full rank, i.e. finite OLS variance).

Exercise 4.9 — Computational Trade-offs

$n = 10^6$, $d = 10^3$. Compare (a) Normal Equations via Gaussian elimination, (b) conjugate gradient, (c) mini-batch GD ($B = 256$). Which would you choose?

Solution. (a) **Normal Equations via Gaussian elimination.** Forming $X^\top X$ costs $O(nd^2)$ and solving the resulting $d \times d$ linear system via Gaussian elimination costs $O(d^3)$. With $n = 10^6$, $d = 10^3$: forming $X^\top X$ is $O(10^6 \cdot 10^6) = O(10^{12})$ operations, and solving is $O(10^9)$ — dominated by the $O(nd^2)$ term. This is a large but one-shot, finite, exact computation (no iteration/convergence tolerance needed); feasible on modern hardware (a few minutes to an hour depending on implementation/hardware), but memory for $X^\top X$ ($d \times d = 10^6$ entries, fine) and for X itself ($nd = 10^9$ entries $\approx$ several GB in float32) must be considered.

(b) **Conjugate gradient (CG) on the Normal Equations.** Each CG iteration requires one matrix-vector product with $X^\top X$ (or equivalently X then $X^\top$, avoiding ever forming $X^\top X$ explicitly), costing $O(nd)$ per iteration = $O(10^9)$. CG converges (to a given tolerance) in at most d iterations in exact arithmetic, but in practice far fewer iterations ($O(\sqrt{\kappa})$ where κ is the condition number of $X^\top X$) suffice for a good approximate solution. If, say, ~ 50 – 200 iterations suffice, total cost is $O(10^{11}$ – $10^{11.3})$ — competitive with or cheaper than (a), and crucially avoids ever materializing the $d \times d$ matrix $X^\top X$ (memory advantage) and avoids numerical issues from squaring the condition number that come with forming $X^\top X$ explicitly.

(c) **Mini-batch gradient descent**, $B = 256$. Each epoch requires $n/B \approx 3900$ mini-batch gradient steps, each costing $O(Bd) = O(256,000)$ operations, so one epoch costs $O(nd) = O(10^9)$, same order as one CG iteration. However, GD-style methods typically need *many* epochs (tens to hundreds, depending on learning rate schedule and conditioning) to converge to a comparable precision, since first-order gradient descent's convergence rate depends unfavorably on the condition number ($O(\kappa)$ or $O(\sqrt{\kappa})$ with momentum, vs. CG's $O(\sqrt{\kappa})$ which is already the best achievable rate

for this problem class). It also introduces optimisation hyperparameters (learning rate, schedule, batch size, momentum) requiring tuning.

Recommendation. For this specific problem — a *convex, exactly-solvable* least-squares problem with $d = 10^3$ manageable and $n = 10^6$ — **conjugate gradient** is generally the best choice: it exploits the exact quadratic structure of the objective (unlike generic mini-batch SGD, which is designed for problems where exact solves are infeasible, e.g. non-convex deep learning or truly enormous d), converges in far fewer effective passes over the data than SGD, needs no learning-rate tuning, and avoids the $O(d^3)$ /large-memory cost and poor numerical conditioning of explicitly forming and factorizing $X^\top X$ as in (a). Mini-batch SGD would be preferred instead if d were far larger (e.g. $d \sim 10^7$, where even $O(nd)$ per full pass is expensive and stochastic estimates become necessary), if the problem were non-convex, or if data arrives in a streaming fashion.